

Enrolment Number

Total No. of printed pages = 04

Winter, 2026

B. Tech 4th Semester Examination

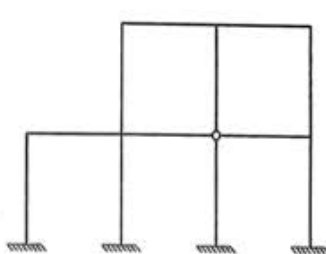
Structural Analysis – Theory

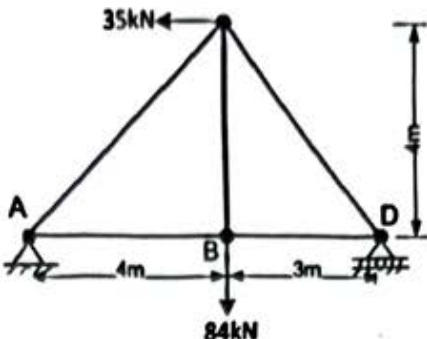
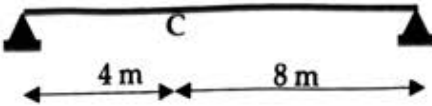
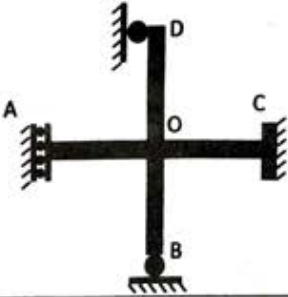
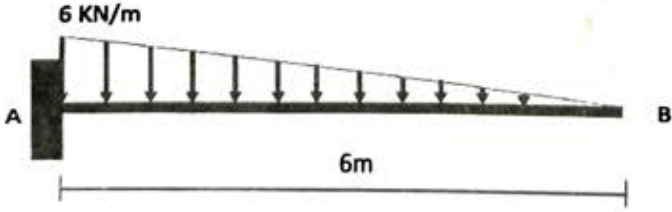
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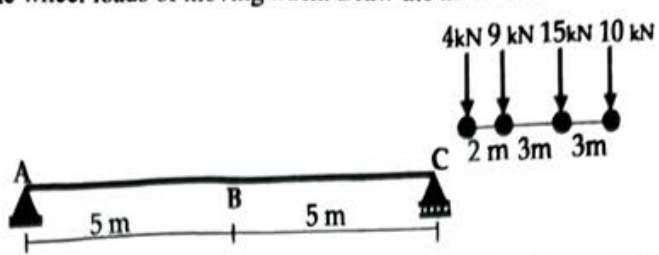
Full Marks: 50

Time: 2 hours

The figure in the margin indicates full marks for the questions.

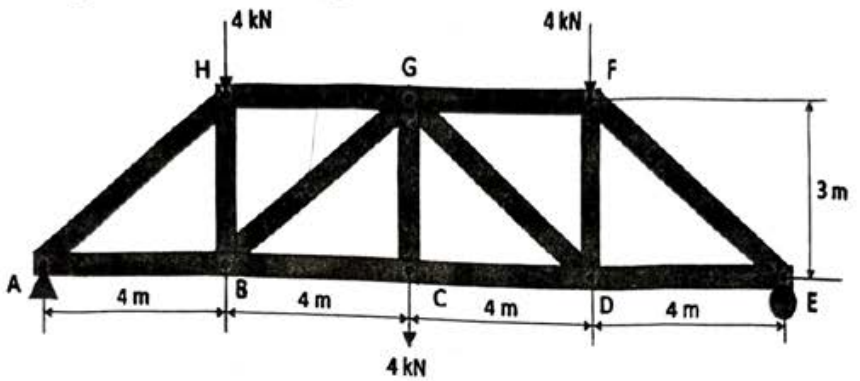
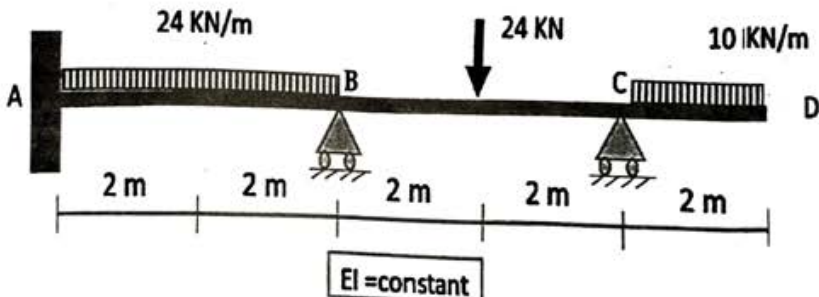
S. No	Questions	Marks
Section 1 [10 Marks]		
1a	1. If the degree of static indeterminacy of a structure is zero, the structure is called <u>statically determinate</u>	1
1b	Influence lines are especially useful for structures subjected to <u>moving</u> loads.	1
1c	What do you mean by stiffness of a member. <u>Ability to resist deformation</u>	1
1d	Define distribution factor. <u>DF = K/EK</u>	1
1e	Rankine's formula combines <u>Crippling</u> and Euler theory.	1
1f	Explain principal of superposition. <u>Total response (deflection, internal force) caused by multiple loads acting together eq. to the algebraic sum of the responses caused by each load acting individually</u>	1
1g	The effective length of a cantilever column is <u>2L</u>	1
1h	Castigliano's theorem is applicable to structures behaving within the <u>elastic</u> range.	1
1i	The fixed end moment for a beam carrying a point load at the midspan is <u>$\frac{WL}{8}$</u> at both ends.	1
1j	The method of joints is mainly used to determine <u>member forces</u>	1
Section 2 [8 Marks]		
Answer any 4 questions [2 marks each]		
2a	Determine the degree of indeterminacy for the rigid-jointed frames indicated in Figures.	2
		

2b	Classify the truss in the figure below as stable, unstable, statically determinate, or static indeterminate. Also calculate the reactions of the given truss. 	2
2c	A three hinged parabolic arch of uniform cross section of span 60m and rise 10m carries a uniformly distributed load of 10 kN/m over the entire span. Calculate the vertical reactions and the horizontal thrust at supports.	2
2d	Draw the influence line diagram for moment at point C of the given beam 	2
2e	Calculate the Distribution factor for each member of the beam shown in figure below. EI and L is constant for all the members. 	2
2f	What is the ratio of the strength of a solid steel column of 200mm diameter to that of a hollow circular steel column of the same cross-sectional area and a wall thickness of 20mm. The two columns have the same length and same boundary condition.	2
Section 3 [12 Marks] Answer any 3 questions [4 marks each]		
3a	Derive the expression for crippling load for columns with one end fixed and other hinged.	4
3b	Determine the maximum deflection of the beam $E = 200 \text{ GPa}$ and $I = 100 \times 10^6 \text{ mm}^4$. Use unit load method. 	4

3c ✓	Determine the maximum positive shear created at point B in the beam shown in Figure due to the wheel loads of moving truck. Draw the influence line for shear. 	4
3d	In the analysis of portal frame ABCD with sway, calculate the Restoring force (R) at the top of the frame. The end column moments are as follows: $M_{AB} = 9.58 \text{ kN-m}$ $M_{CD} = -15.00 \text{ kN-m}$ $M_{BA} = 19.34 \text{ kN-m}$ $M_{DC} = 0 \text{ kN-m}$ The length of column AB and CD are 10 m and 15 m respectively.	4

Section 4 [20 Marks]

Answer any 2 questions [10 marks each]

4a ✓	Calculate the forces in each member of the truss using method of joints. Support A is hinged and E is a roller support. 	10
4b ✓	Analyse the beam shown in Figure using Slope deflection method if support B sinks by 20 mm. Draw the bending moment diagrams for the beam. Take $EI = 2 \times 10^4 \text{ kN-m}^2$ 	10

4c

✕

Analyse the frame shown using **Moment distribution method**. Draw the bending moment diagrams for the frame. EI is constant.

10

